

WFLOW-04: Advanced Notification Routing

Series: WFLOW — Workflows and Alert Notifications | **Notebook:** 4 of 10 | **Created:** January 2026 | **Last Updated:** 09/24/2026

Intelligent Alert Routing

Not all alerts should go to everyone. This notebook covers conditional routing based on severity, team ownership, time of day, and escalation patterns.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription

Requirement	Details
Permissions	<code>automation:workflows:write</code>
Prior Knowledge	WFLOW-01 through WFLOW-03
Connections	Slack and/or Teams connections configured

Sprint 1.337 (April 2026): Smartscape Ownership for Routing

SaaS 1.337 made ownership available on Smartscape nodes, for use from **Workflows actions**: *"Ownership information is now available in Smartscape and for Smartscape nodes. This lets you use Workflows actions (such as `get_owners`) to send automated and targeted notifications to your teams"* ([What's new in SaaS 1.337 \(DT docs\)](#)). It is not a DQL field. An earlier revision of this notebook read

```
getNodeField(affected_entity_ids[0], "ownership.team") ; that returns null on every problem, because no Smartscape node model has an ownership.* field and getNodeField does not resolve a classic entity ID.
```

Two documented routes:

- **Route on ownership with the Ownership app's *Get owners* action.** Add a *Get owners* task after the Problem trigger. Its *Entity ids* input takes "A Jinja expression, for example, `{{ event()["dt.entity.host"] }}`"; its output carries `slackChannels`, `msTeams`, `email` and `jira` contact lists per team ([Actions for Ownership \(DT docs\)](#)). Feed those into the notification task (for example `{{ result("get_owners").slackChannels }}`) instead of hard-coding team names in the workflow.
- **Or route on `primary_tags.team`** in the trigger's custom filter or a task condition. *"Dynatrace enriches all derived signals (service metrics, Davis events, and problems) with the same tags."* ([Primary Grail fields and tags \(DT docs\)](#)). The matcher is `primary_tags.team == "payments"`. Verify your entities carry the tag first: on the validation tenant `primary_tags.team` was null on every problem (09/24/2026).

FAQ-21 §5 covers ownership coverage and the Problem fields mapping in depth.

Sprint 1.337 (Dynatrace API): `metadata` removed from `GET /events`

The Events API endpoint `GET /events` no longer returns the `metadata` property in event query results or individual event responses. Workflows that triggered off `dt.davis.events` and parsed event-level `metadata` for additional context need to reach into `event.*`

semantic fields directly. Update any HTTP-action JSON parsing to drop the `metadata` field reference.

1. Routing Strategies

 **Migrating from alerting profiles? Duration suppression maps onto the trigger's Minimum duration option.** An alerting profile can delay notification until a problem has been open longer than *N* minutes (`delayInMinutes`) — a common way to suppress transient blips. The problem trigger's **Minimum duration** option (renamed from **Delay** in 08/2026) postpones *"the trigger until the problem has been open for at least the configured duration"* — 5, 10, 15, 30, 60, 120, 240, 1440, or 10080 minutes, evaluated on `dt.duration_marker` ([Problem and event triggers \(DT docs\)](#)). Since its 09/07/2026 rewrite the [alert-notification upgrade guide \(DT docs\)](#) agrees: *"The delay, update, and severity capabilities described in this guide exist only on the workflow trigger."* Earlier versions of that guide said there was no alternative; that line is gone.

The values are fixed, so a delay that is not one of those nine has to be rounded — up suppresses a little more, down pages a little sooner. Treat a long delay as evidence the alert was the wrong *shape* rather than merely delayed: a profile suppressing 30 minutes of a firing condition is usually describing a burn-rate concern, which belongs in an SLO burn-rate alert (**SLO-04**) or a Davis anomaly detector (**AIOPS-02**). Inventory `delayInMinutes` across every profile and choose a Minimum duration for each; no special migration wave is needed. See **MZ2POL-09** §6.1.

Why Route Alerts?

Problem	Impact	Solution
Alert fatigue	Teams ignore alerts	Route only relevant alerts
Slow response	Wrong team notified	Route to owners
Off-hours noise	Sleep disruption	Time-based routing
Missed escalation	Unacknowledged alerts	Auto-escalation

Routing Dimensions

Dimension	Route Based On	Example
Severity	Problem severity level	Critical → PagerDuty, Low → Slack
Team	Entity tags, or Smartscape ownership	<code>team:checkout</code> → <code>#checkout-alerts</code>
Service	Service name or entity ID	Payment service → payments team
Time	Hour of day, day of week	Weekends → on-call only
Environment	prod/staging/dev tags	Prod → immediate, Dev → daily digest

2. Conditional Expressions

Conditions control which tasks execute. They use Jinja expressions returning boolean values.

Defining Conditions

```
conditions:
- name: is_critical
  expression: '{{ (event().get("event.severity") | int(5)) <= 1 }}'

- name: is_production
  expression: '{{ "prod" in event()["management_zones"] }}'

- name: is_checkout_team
  expression: '{{ "team:checkout" in event().get("tags", []) }}'
```

Using Conditions on Tasks

```
tasks:
  - name: pagerduty_alert
    type: dynatrace.pagerduty:create-incident
    conditions:
      - is_critical
      - is_production
    # Task runs only if BOTH conditions are true (AND logic)
```

Negating Conditions

```
tasks:
  - name: slack_only_for_non_critical
    type: dynatrace.slack:message
    conditions:
      - not is_critical
    # Task runs when is_critical is FALSE
```

Complex Expressions

```
# AND within expression
{{ (event().get("event.severity") | int(5)) <= 1 and "prod" in event()
["management_zones"] }}

# OR within expression
{{ (event().get("event.severity") | int(5)) <= 2 }}

# Check if field exists
{{ event().get("root_cause_entity_id") is not none }}
```

3. Routing by Severity

Severity-Based Routing Pattern

The problem record carries severity as `event.severity`, a 1–5 scale (1 = most severe). There is no `CRITICAL / HIGH / MEDIUM / LOW` string on it, so conditions comparing against those strings are false for every problem. The tiers below use the scale:

<code>event.severity</code>	Actions
1 (Critical)	PagerDuty + Slack #urgent + Email
2 (High)	Slack #alerts + Email
3 (Medium)	Slack #alerts
4 (Low)	Slack #alerts (business hours only)

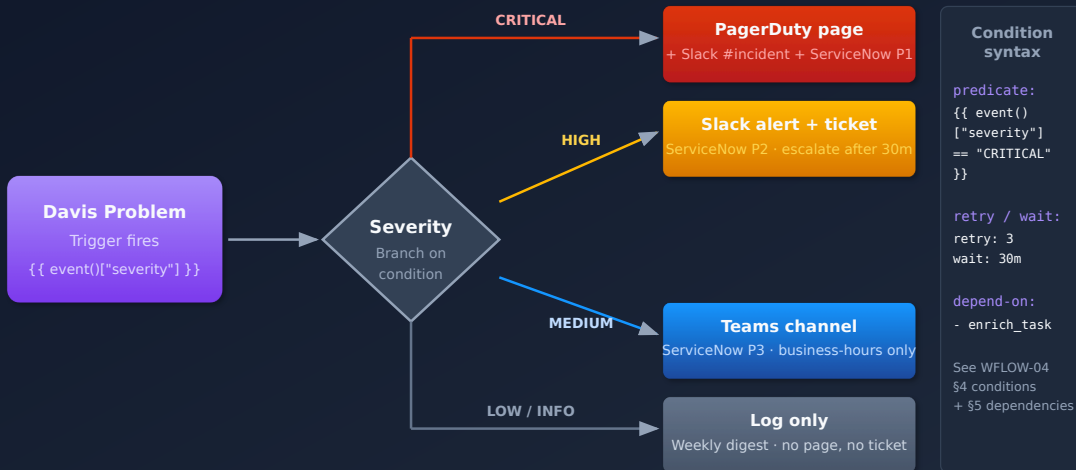
Breaking — SaaS 1.348 (pre-release; staged tenant rollout planned from 09/22/2026): severity is no longer defaulted. *"Davis events and problems no longer default `event.severity` to 3."* A severity filter that was matching the default stops matching once 1.348 reaches your tenant: *"Workflows with a Davis event/problem trigger that filter on `event.severity=3` expecting it to be defaulted, might need to be changed to filter for Any severity to keep the alerts."* Before relying on severity tiers, check which event sources actually **set** a severity (on the validation tenant, every problem in 30 days carried the default 3 bar one). For sources that do not, route on `event.category`, ownership or tags instead, or restore a default with a Davis event OpenPipeline processor. Until 1.348 reaches your tenant, the defaulting behavior still applies and the conditions below keep working. The `int(5)` in them treats a missing severity as the least severe tier, so an unset severity never pages. **FAQ-21 §6** carries the same note.

Sources: [What's new in Dynatrace SaaS 1.348 \(DT docs\)](#) — pre-release, read 09/24/2026.

Dictionary: `event.severity` (`experimental`), read 09/24/2026.

Severity-Based Notification Routing

Single workflow, branched by Davis problem severity into 4 distinct notification paths



Workflow Configuration

```
conditions:
  - name: is_critical
    expression: '{{ (event().get("event.severity") | int(5)) &lt;= 1 }}'
  - name: is_high
    expression: '{{ (event().get("event.severity") | int(5)) == 2 }}'
  - name: is_medium_or_low
    expression: '{{ (event().get("event.severity") | int(5)) in [3, 4] }}'

tasks:
  # CRITICAL: Page on-call + Slack urgent + Email
  - name: pagerduty_critical
    type: dynatrace.pagerduty:create-incident
    conditions: [is_critical]
    input:
      connection: pagerduty-prod
      severity: critical
      summary: "{{ event()['event.name'] }}"

  - name: slack_urgent
    type: dynatrace.slack:message
    conditions: [is_critical]
    input:
      connection: slack-production
      channel: "#alerts-urgent"
      message: ":rotating_light: *CRITICAL* {{ event()['event.name'] }}"

  # HIGH: Slack alerts + Email
  - name: slack_high
    type: dynatrace.slack:message
    conditions: [is_high]
    input:
      channel: "#alerts-production"
      message: ":warning: *HIGH* {{ event()['event.name'] }}"

  # MEDIUM/LOW: Slack only
  - name: slack_low
    type: dynatrace.slack:message
    conditions: [is_medium_or_low]
    input:
      channel: "#alerts-production"
      message: ":information_source: *{{ event()['event.category'] }}* {{ event()['event.name'] }}"
```

Update — unified `event.severity` field (2026). Dynatrace now exposes a standardized `event.severity` value as an **integer 1–5** (1=Critical, 2=High, 3=Medium, 4=Low, 5=Informational), aligned to the ITIL severity model. It propagates from the constituent alerts up to the parent problem, which makes severity a first-class, consistent routing dimension across alerts, the problem feed, and Workflows.

Integer	Tier
1	Critical
2	High
3	Medium
4	Low
5	Informational

The conditions in this section key on this numeric field. Earlier revisions compared against `"CRITICAL"` / `"HIGH"` / `"MEDIUM"` / `"LOW"` ; the problem record carries no such strings, so those conditions were false for every problem. The field is `experimental` in the semantic dictionary and arrives as a string (`"3"`) on the validation tenant, hence the `| int` conversion. In DQL (problem feed, reporting) the field is queried directly as `event.severity` — see **AIOPS-03 §5** for the severity rollup pattern.

4. Routing by Team/Service

Tag-Based Team Routing

Entities tagged with `team:checkout` , `team:payments` , etc.

```

conditions:
  - name: is_checkout_team
    expression: |
      {% set tags = event().get("tags", []) %}
      {{ "team:checkout" in tags or "team:cart" in tags }}

  - name: is_payments_team
    expression: '{{ "team:payments" in event().get("tags", []) }}'

  - name: is_platform_team
    expression: '{{ "team:platform" in event().get("tags", []) }}'

tasks:
  - name: notify_checkout
    type: dynatrace.slack:message
    conditions: [is_checkout_team]
    input:
      channel: "#checkout-alerts"
      message: '{{ event()["event.name"] }}'

  - name: notify_payments
    type: dynatrace.slack:message
    conditions: [is_payments_team]
    input:
      channel: "#payments-alerts"
      message: '{{ event()["event.name"] }}'

  - name: notify_platform
    type: dynatrace.slack:message
    conditions: [is_platform_team]
    input:
      channel: "#platform-alerts"
      message: '{{ event()["event.name"] }}'

```

Management Zone Routing (legacy — do not build new workflows on this)

 **This pattern breaks silently when Management Zones are retired.** `event()` `["management_zones"]` still resolves after the zones are deleted — to an **empty array** — so every condition below evaluates false and the workflow stops notifying without raising an error. There is no Management Zone filter on the problem trigger itself (WFLOW-02 §2).

Use the tag-based pattern above instead. If a region is the routing dimension, tag the entities (`region:us-east`) or use Smartscape ownership rather than reading the MZ array. Teams actively migrating off MZs should read MZ2POL-01 §5.

Retained for reference, and only valid while your Management Zones still exist:

```
conditions:
  - name: is_us_region
    expression: '{{ "US-East" in event()["management_zones"] or "US-West" in event()["management_zones"] }}'

  - name: is_eu_region
    expression: '{{ "EU-West" in event()["management_zones"] }}'

tasks:
  - name: notify_us_team
    conditions: [is_us_region]
    input:
      channel: "#us-oncall"

  - name: notify_eu_team
    conditions: [is_eu_region]
    input:
      channel: "#eu-oncall"
```

Dynamic Channel Selection

Use JavaScript to determine the channel dynamically:

```
import { execution } from '@dynatrace-sdk/automation-utils';

export default async function () {
  const ev = (await execution()).params.event; // trigger payload
  const tags = ev.entity_tags || [];           // "key:value" strings

  // Find team tag
  const teamTag = tags.find(t => t.startsWith('team:'));
  const team = teamTag ? teamTag.split(':')[1] : 'platform';

  return {
    channel: `#${team}-alerts`,
    team: team
  };
}
```

5. Time-Based Routing

Business Hours vs Off-Hours

```
conditions:
- name: is_business_hours
  expression: |
    {% set hour = now().hour %}
    {% set weekday = now().weekday() %}
    {{ weekday < 5 and hour >= 9 and hour < 17 }}

- name: is_off_hours
  expression: |
    {% set hour = now().hour %}
    {% set weekday = now().weekday() %}
    {{ weekday >= 5 or hour < 9 or hour >= 17 }}

tasks:
# Business hours: Slack channel
- name: slack_business_hours
  type: dynatrace.slack:message
  conditions: [is_business_hours]
  input:
    channel: "#alerts-production"
    message: "{{ event()['event.name'] }}"

# Off-hours: PagerDuty for CRITICAL only
- name: pagerduty_off_hours
  type: dynatrace.pagerduty:create-incident
  conditions: [is_off_hours, is_critical]
  input:
    connection: pagerduty-prod
    summary: "[Off-Hours] {{ event()['event.name'] }}"

# Off-hours non-critical: Queue for morning
- name: slack_queue
  type: dynatrace.slack:message
  conditions: [is_off_hours, not is_critical]
  input:
    channel: "#alerts-queue"
    message: ":moon: *Queued for review* {{ event()['event.name'] }}"
```

Weekend-Only Routing

```
conditions:
  - name: is_weekend
    expression: '{{ now().weekday() >= 5 }}'

tasks:
  - name: weekend_oncall
    conditions: [is_weekend, is_critical]
    input:
      # Page weekend on-call rotation: a connection that holds that rotation's
      # routing key. Credentials live in connections, not in expressions.
      connection: pagerduty-weekend-oncall
```

6. Escalation Patterns

Timed Escalation

Escalate if not acknowledged within a time window.

```

tasks:
  # Step 1: Initial notification
  - name: initial_slack
    type: dynatrace.slack:message
    input:
      channel: "#alerts-production"
      message: ":warning: {{ event()['event.name'] }} - Acknowledge within 15 min"

  # Step 2: Wait 15 minutes
  - name: wait_for_ack
    type: dynatrace.automations:wait
    dependsOn: [initial_slack]
    input:
      duration: "15m"

  # Step 3: Check if still open (event.status is ACTIVE or CLOSED – never OPEN)
  - name: check_problem_status
    type: dynatrace.automations:run-javascript
    dependsOn: [wait_for_ack]
    input:
      script: |
        import { execution } from '@dynatrace-sdk/automation-utils';
        import { queryExecutionClient } from '@dynatrace-sdk/client-query';

        export default async function () {
          const ev = (await execution()).params.event;    // snapshot from trigger time
          // Re-query: the trigger payload is 15 minutes old by now
          const q = await queryExecutionClient.queryExecute({ body: { query:
            `fetch dt.davis.problems, from:-2h | filter display_id ==
            "${ev['display_id']}" | sort timestamp desc | limit 1 | fields event.status` }
          });
          return { escalate: q.result.records[0]?['event.status'] ===
            'ACTIVE' };
        }

  # Step 4: Escalate to PagerDuty
  - name: escalate_pagerduty
    type: dynatrace.pagerduty:create-incident
    dependsOn: [check_problem_status]
    conditions:
      - '{{ result("check_problem_status").escalate }}'
    input:
      severity: high

```

```
summary: "[ESCALATED] {{ event()['event.name'] }} - No acknowledgment in 15 min"
```

Multi-Tier Escalation

```
0 min → Slack channel notification  
15 min → Email to team lead  
30 min → PagerDuty to on-call  
60 min → PagerDuty to manager
```

7. Multi-Channel Strategy

Recommended Channel Matrix

event.severity	Environment	Channel(s)
1 (Critical)	Production	PagerDuty + Slack urgent + Email
1 (Critical)	Staging	Slack alerts
2 (High)	Production	Slack alerts + Email
2 (High)	Staging	Slack alerts
3 (Medium)	Production	Slack alerts
3 (Medium)	Staging	Slack (daily digest)
4 (Low)	Any	Slack (weekly digest)

Complete Multi-Channel Workflow

```
conditions:
  - name: is_critical
    expression: '{{ (event().get("event.severity") | int(5)) &lt;= 1 }}'
  - name: is_high_or_above
    expression: '{{ (event().get("event.severity") | int(5)) &lt;= 2 }}'
  - name: is_production
    expression: '{{ "env:prod" in event().get("tags", []) }}'
  - name: is_business_hours
    expression: '{{ now().weekday() &lt; 5 and now().hour &gt;= 9 and now().hour &lt; 17 }}'

tasks:
  # PagerDuty: Critical + Production
  - name: pagerduty
    type: dynatrace.pagerduty:create-incident
    conditions: [is_critical, is_production]

  # Slack Urgent: Critical + Production
  - name: slack_urgent
    type: dynatrace.slack:message
    conditions: [is_critical, is_production]
    input:
      channel: "#alerts-urgent"

  # Slack Standard: High or above + Production
  - name: slack_standard
    type: dynatrace.slack:message
    conditions: [is_high_or_above, is_production]
    input:
      channel: "#alerts-production"

  # Email: Critical or (High + Production + Business Hours)
  - name: email_alert
    type: dynatrace.email:send-email
    conditions:
      - '{{ (event().get("event.severity") | int(5)) &lt;= 1 or ((event().get("event.severity") | int(5)) == 2 and "env:prod" in event().get("tags", [])) }}'
```

Query Workflow Routing Effectiveness

```
// Workflow executions by outcome
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`,
// and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in
// any spelling – those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to
// `dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by
// `event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION ·
//   WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type – e.g. run-javascript,
//   send-email,
//                                       snow-search-incidents)
//   .state                             (was task.status / execution.status)
//                                       values RUNNING · SUCCESS · ERROR ·
//   DISCARDED · SKIPPED
//                                       – note "FAILED" is NOT a value; it
//   is ERROR
//   .state.is_final                    true only on terminal records –
//   filter on it, otherwise a
//                                       single execution is counted once as
//   RUNNING and again as
//                                       SUCCESS/ERROR
//   .state_info                        (was task.error) · duration (was
//   task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT"
//   | limit 1
//   fetch dt.system.events, from:-7d
//   | filter event.kind == "WORKFLOW_EVENT"
//   | filter event.type == "WORKFLOW_EXECUTION"
//   | filter dt.automation_engine.state.is_final == true
//   | summarize executions = count(), by:{dt.automation_engine.state}
//   | sort executions desc
```

```

// Task execution distribution by notification channel
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`,
and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in
any spelling – those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to
`dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by
`event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION ·
WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type – e.g. run-javascript,
send-email,
//                                       snow-search-incidents)
//   .state                             (was task.status / execution.status)
//                                       values RUNNING · SUCCESS · ERROR ·
DISCARDED · SKIPPED
//                                       – note "FAILED" is NOT a value; it
is ERROR
//   .state.is_final                    true only on terminal records –
filter on it, otherwise a
//                                       single execution is counted once as
RUNNING and again as
//                                       SUCCESS/ERROR
//   .state_info                        (was task.error) · duration (was
task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT"
| limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "ACTION_EXECUTION"
| filter dt.automation_engine.state.is_final == true
| summarize executions = count(), by:{dt.automation_engine.action.app,
dt.automation_engine.action.function}
| sort executions desc
| limit 20

```

```
// Skipped tasks (conditions not met)
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`,
and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in
any spelling – those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to
`dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by
`event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION ·
WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type – e.g. run-javascript,
send-email,
//                                       snow-search-incidents)
//   .state                             (was task.status / execution.status)
//                                       values RUNNING · SUCCESS · ERROR ·
DISCARDED · SKIPPED
//                                       – note "FAILED" is NOT a value; it
is ERROR
//   .state.is_final                    true only on terminal records –
filter on it, otherwise a
//                                       single execution is counted once as
RUNNING and again as
//                                       SUCCESS/ERROR
//   .state_info                        (was task.error) · duration (was
task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT"
| limit 1
fetch dt.system.events, from:-24h
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "TASK_EXECUTION"
| filter in(dt.automation_engine.state, {"SKIPPED", "DISCARDED"})
| summarize skipped = count(), by:{dt.automation_engine.workflow.title,
dt.automation_engine.task.name, dt.automation_engine.state}
| sort skipped desc
| limit 25
```

Next Steps

With routing configured, integrate with incident management:

Recommended Path

1. **WFLOW-05: PagerDuty & ServiceNow** - Create incidents automatically
2. **WFLOW-06: Custom Templates** - Rich message formatting
3. **WFLOW-07: Problem-Triggered Remediation** - Auto-remediation

Key Takeaways

- **Conditions** control task execution using Jinja expressions
 - **Severity routing** ensures appropriate response levels
 - **Team routing** uses entity tags or Smartscape ownership — not management zones
 - **Time-based routing** reduces off-hours noise
 - **Escalation patterns** prevent missed alerts
 - Test conditions with On-Demand trigger before production
-

Summary

In this notebook, you learned:

- Why and when to route alerts conditionally
 - Conditional expression syntax and operators
 - Severity-based routing patterns
 - Team routing by entity tag, and why management-zone routing is now legacy
 - Time-based (business hours) routing
 - Escalation with wait and check patterns
 - Multi-channel notification strategy
-

References

- [Workflow reference / Jinja expressions + conditions \(DT docs\)](#)
- [Workflow actions umbrella \(DT docs\)](#)
- [Notification actions umbrella \(DT docs\)](#)

- [Davis Problems app \(DT docs\)](#)
 - [Alerting and notifications umbrella \(DT docs\)](#)
 - [Upgrade guide — alerting and notifications \(DT docs\)](#)
-

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